

1. If you have 10,000 examples, how would you split the train/dev/test set? Choose the best option. 1 point
- ☒ 60% train, 20% dev, 20% test.
 - ☐ 33% train, 33% dev, 33% test.
 - ☐ 98% train, 1% dev, 1% test.
2. The dev and test set should: 1 point
- ☐ Be identical to each other (same (x,y) pairs)
 - ☒ Come from the same distribution
 - ☐ Come from different distributions
 - ☐ Have the same number of examples
3. If your Neural Network model seems to have high variance, what of the following would be promising things to try? 1 point
- ☐ Make the Neural Network deeper
 - ☐ Get more test data
 - ☒ Get more training data
 - ☐ Increase the number of units in each hidden layer
 - ☒ Add regularization
4. You are working on an automated check-out kiosk for a supermarket and are building a classifier for apples, bananas, and oranges. 1 point
- Suppose your classifier obtains a training set error of 19% and a development set error of 21%.
- Which of the following is the most promising strategy to improve your classifier? (Assume the human error is approximately 0%)
- ☐ Get more training data.
 - ☒ Use a bigger network.
 - ☐ Increase the regularization parameter lambda.
5. Which of the following are regularization techniques? 1 point
- ☒ Dropout.
 - ☐ Gradient Checking.
 - ☐ Increase the number of layers of the network.
 - ☒ Weight decay.
6. True or False: To reduce high variance, the regularization hyperparameter lambda must be increased. 1 point
- ☒ True
 - ☐ False
7. Which of the following are true about dropout? 1 point
- ☐ It helps to reduce the bias of a model.
 - ☐ In practice, it eliminates units of each layer with a probability of keep_prob.
 - ☒ It helps to reduce overfitting.
 - ☒ In practice, it eliminates units of each layer with a probability of $1 - \text{keep_prob}$.
8. Increasing the parameter keep_prob from (say) 0.5 to 0.6 will likely cause the following: (Check the two that apply) 1 point
- ☐ Increasing the regularization effect
 - ☒ Reducing the regularization effect
 - ☐ Causing the neural network to end up with a higher training set error
 - ☒ Causing the neural network to end up with a lower training set error
9. Which of these techniques are useful for reducing variance (reducing overfitting)? (Check all that apply.) 1 point
- ☐ Gradient Checking
 - ☐ Xavier initialization
 - ☒ Dropout
 - ☒ Data augmentation
 - ☐ Exploding gradient
 - ☒ L2 regularization
 - ☐ Vanishing gradient
10. Suppose that a model uses, as one feature, the total number of kilometers walked by a person during a year, and another feature is the height of the person in meters. What is the most likely effect of normalization of the input data? 1 point
- ☐ It won't have any positive or negative effects.
 - ☐ It will make the data easier to visualize.
 - ☒ It will make the training faster.
 - ☐ It will increase the variance of the model.